Rockchip Gpio Output Clocks

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Preface

Overview

Product Version

Chip Name	Kernel Version
RK3399	Linux 4.4 and above
RK1808	Linux 4.4 and above
RK3328	Linux 4.4 and above
RK3308	Linux 4.4 and above
RV1126	Linux 4.19 and above
PX30	Linux 4.4 and above
RK3566/8	Linux 4.19 and above
RK3588	Linux 5.10
RK3528	Linux 4.19
RK3576	Linux 6.1
RK2118	RTT
RV1126B/RV1126B-P	Linux 6.1
RK2116	RTT
RK1820	RTT
RK3538/9	linux6.1

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Zhang Qing	2021-09-06	Initial version
V1.1.0	Zhang Qing	2021-12-29	Added RK3588 support
V1.2.0	Zhang Qing	2023-02-27	Added RK3528 support
V1.3.0	Zhang Qing	2024-04-27	Added RK3576 & RK2118 support
V1.4.0	Zhang Qing	2025-04-16	Added RV1126B/RV1126B-P support
V1.5.0	Zhang Qing	2025-07-18	Added RK2116/RK1820 support
V1.6.0	Zhang Qing	2026-01-09	Added RK3538/9 support

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1. Concept Introduction

1.1 Concept Explanation

With the increase in peripheral devices and cost constraints, many peripherals would like to use the SOC's IO to output specific clock frequencies to replace external crystal oscillators, thereby reducing the cost of peripheral circuits.

1.2 Main Configuration

The SOC's IO output CLK is predicated on the IOMUX of this IO having a CLK output function.

1. IOMUX

The IOMUX of this IO must be configured to output CLK functionality.

2. CLK Frequency

CLK frequency needs to be configured. Since it is meant to replace an external crystal, the best signal quality is achieved when this CLK outputs 24M and is bypassed from the SOC's crystal. If the signal is derived from a PLL divider, the signal quality is slightly worse, and whether it can meet the peripheral requirements depends on the peripheral's reference documentation.

3. IO Drive Capability

The waveforms tested on some IOs are not very appealing, and it may be necessary to adjust the IO's drive capability.

2. Correspondence Relationship

2.1 GPIO and CLK Hardware Correspondence

Chipset	clk name	gpio name	Supported Frequencies
RK3288	SCLK_TESTOUT	GPIO0_C1	24M, 32K
RK3328	SCLK_TESTOUT	GPIO1_A6	24M, 32K
PX30	SCLK_TEST_OUT	GPIO1_D6\GPIO0_B5	24M, 32K
RK3308	SCLK_TESTOUT	GPIO0_A4	24M, 32K
RV1108	SCLK_TESTOUT	GPIO0_A6	24M, 32K
RV1126	SCLK_TESTOUT	GPIO1_A4	24M, 32K
RK3368	SCLK_TESTOUT	GPIO0_B0	24M, 32K
RK3399	SCLK_TESTOUT	GPIO2_D1\GPIO0_B0	24M, 32K
RK3566/8	CLK_WIFI	GPIO0_A0	24M
RK3566/8	CLK_MAC0_OUT	GPIO2_C1	24M, 25M, 50M, 125M
RK3566/8	CLK_MAC1_OUT	GPIO3_B0	24M, 25M, 50M, 125M
RK3566/8	CLK_MAC1_OUT	GPIO4_B3	24M, 25M, 50M, 125M
RK3566/8	CLK_CIF_OUT	GPIO4_C0	24M, 27M, 37.125M
RK3566/8	CLK_CAM0_OUT	GPIO4_A7	24M, 27M, 37.125M
RK3588	REFCLKOUT	GPIO0_A0	24M
RK3588	CLK_GMAC_50M	GPIO4_C3	50M, 125M
RK3588	REFCLK25M_ETH0	GPIO4_B0	25M, 50M
RK3588	CLK_DEEPSLOW	GPIO2_C5	32K
RK3588	CLK_MIPI_CAMARAOUT_M0	GPIO4_B1	24M, 27M, 37.125M
RK3588	CLK_MIPI_CAMARAOUT_M1	GPIO1_B6	24M, 27M, 37.125M
RK3588	CLK_MIPI_CAMARAOUT_M2	GPIO1_B7	24M, 27M, 37.125M
RK3588	CLK_MIPI_CAMARAOUT_M3	GPIO1_D6	24M, 27M, 37.125M
RK3588	CLK_MIPI_CAMARAOUT_M4	GPIO1_D7	24M, 27M, 37.125M
RK3588	CLK_CIF_OUT	GPIO4_B4	24M, 27M, 37.125M
RK3588	REFCLK25M_ETH1	GPIO3_A6	25M, 50M
RK3528	CLK_REFOUT	GPIO0_A1	24M
RK3528	CLK_REFOUT	GPIO3_C3	24M
RK3528	CLK_DEEPSLOW	GPIO3_C3	32K
RK3528	CLK_DEEPSLOW	GPIO1_C3	32K
RK3528	CLK_TESTOUT	GPIO2_A5	Depend on actual configuration

Chipset	clk name	gpio name	Supported Frequencies
RK3576	REF_CLK0_OUT	GPIO0_A0	12M, 24M, 27M, 37.125M
RK3576	REF_CLK1_OUT	GPIO0_B4	12M, 24M, 27M, 37.125M
RK3576	REF_CLK2_OUT	GPIO0_B5	12M, 24M, 27M, 37.125M
RK3576	CLK_32K_OUT	GPIO0_A2	32K
RK3576	ETH_CLK0_25M_OUT_M0\M1	GPIO3_A4\GPIO2_D7	25M, 27M, 50M, 125M
RK3576	ETH_CLK1_25M_OUT_M0\M1	GPIO2_D6\GPIO1_D5	25M, 27M, 50M, 125M
RK3576	VI_CIF_CLKOUT	GPIO3_A2	24M, 27M, 37.125M
RK3576	CAM_CLK0_OUT_M0\M1	GPIO3_D7\GPIO2_D2	24M, 25M, 26M, 27M, 37.125M, 50M
RK3576	CAM_CLK1_OUT_M0\M1	GPIO4_A0\GPIO2_D6	24M, 25M, 26M, 27M, 37.125M, 50M
RK3576	CAM_CLK2_OUT_M0\M1	GPIO4_A1\GPIO2_D7	24M, 25M, 26M, 27M, 37.125M, 50M
RK3576	UFS_REFCLK	GPIO4_D1	24M, 26M
RK3576	CLK_TESTOUT	GPIO2_A5	Depend on actual configuration
RK2118	REF_CLK0_OUT	GPIO3_B1	24M, 24.576M, 40M, 49.152M, 50M
RK2118	REF_CLK1_OUT	GPIO0_A3	24M, 24.576M, 40M, 49.152M, 50M
RK2118	ETH_CLK_25M_OUT	GPIO0_A2	25M, 50M
RK2118	OSC_CLK_OUT	GPIO0_B3	Crystal bypass
RK2118	CLK_32K	GPIO0_A4	32K
RK2118	CLK_TESTOUT	GPIO3_C6	Depend on actual configuration
RV1126B/RV1126B-P	SAI0_MCLK_M0\M1	GPIO7_A2\GPIO6_A4	12.288M, 24M, 24.576M, 49.152M
RV1126B/RV1126B-P	SAI1_MCLK_M0\M1	GPIO1_B0\GPIO4_A3	12.288M, 24M, 24.576M, 49.152M
RV1126B/RV1126B-P	SAI2_MCLK_M0\M1	GPIO3_B6\GPIO5_A3	12.288M, 24M, 24.576M, 49.152M
RV1126B/RV1126B-P	CAM_CLK0_OUT	GPIO4_B1	24M, 25M, 26M, 27M, 37.125M, 50M

Chipset	clk name	gpio name	Supported Frequencies
RV1126B/RV1126B-P	CAM_CLK1_OUT	GPIO4_B0	24M, 25M, 26M, 27M, 37.125M, 50M
RV1126B/RV1126B-P	CAM_CLK2_OUT	GPIO4_A1	24M, 25M, 26M, 27M, 37.125M, 50M
RV1126B/RV1126B-P	CAM_CLK3_OUT	GPIO4_A0	24M, 25M, 26M, 27M, 37.125M, 50M
RV1126B/RV1126B-P	VI_CIF_CLKOUT_M0\M1	GPIO6_C2\GPIO5_C5	24M, 25M, 26M, 27M, 37.125M, 50M
RV1126B/RV1126B-P	ETH_CLK_25M_OUT_M0\M1	GPIO6_C1\GPIO5_C1	24M, 25M, 26M, 27M, 37.125M, 50M
RV1126B/RV1126B-P	CLK_32K_OUT	GPIO0_A2	32K
RV1126B/RV1126B-P	CLK_REFOUT	GPIO0_A0	24M
RK2116	ETH_CLK_25M_OUT	GPIO0_A2	25M, 50M
RK2116	OSC_CLK_OUT	GPIO0_A4	Crystal bypass
RK2116	CLK_32K	GPIO0_A1	32K
RK2116	CLK_TESTOUT	GPIO2_A3	Depend on actual configuration
RK1820	CLK_TESTOUT0	GPIO0_A0	Depend on actual configuration
RK1820	CLK_TESTOUT1	GPIO0_A2	Depend on actual configuration
RK3538/9	REF_CLK0_OUT	GPIO0_A1	12M, 24M, 27M, 37.125M
RK3538/9	REF_CLK1_OUT	GPIO5_C2	12M, 24M, 27M, 37.125M
RK3538/9	CLK_32K_OUT	GPIO3_C2\GPIO0_D4	32K
RK3538/9	ETH_CLK0_25M_OUT_M0\M1	GPIO0_D5\GPIO0_B4	25M, 27M, 50M, 125M
RK3538/9	CLK_TESTOUT	GPIO2_A5\GPIO0_A1	Depend on actual configuration

3. Chip Example

3.1 IO Command Examples

3.1.1 RK3288

1. Test_clk outputs 24M, test point GPIO0_C1

```
io -4 0xff760068 0x0f000000 # test clk div set 1
io -4 0xff760170 0x80000000 # enable test clk
io -4 0xff7601e8 0x0f000800 # test clk select 24m
io -4 0xff73008c 0x000c0005 # gpio0_c1 iomux select testclk
```

3.1.2 RK3328

1. Test_clk outputs 24M, test point GPIO1_A6

```
io -4 0xff440108 0x000f0000 # set test clk div to 1
io -4 0xff440200 0x02000000 # enable test clk
io -4 0xff100010 0x30002000 # gpio1_a6 iomux select testclk
io -4 0xff440084 0x1f001700 # select 24m for test clk
```

3.1.3 PX30

1. Test_clk outputs 24M, test points GPIO1_D6 or GPIO0_B5

```
io -4 0xff2b0240 0x80000000 # enable test clk
io -4 0xff2b01e4 0x1f1f0007 # test clk select 24m and div set 1
io -4 0xff14001c 0x0f000300 # gpio1_d6 iomux select testclk
io -4 0xff010004 0x0c000800 # gpio0_b5 iomux select testclk
```

3.1.4 RK3308

1. Test_clk outputs 24M, test point GPIO0_A4

```
io -4 0xff500224 0x1fff1700 # test clk select 24m and div set 1
io -4 0xff500310 0x02000000 # enable test clk
io -4 0xff000000 0x03000100 # gpio0_a4 iomux select testclk
```

3.1.5 RV110X

1. Test_clk outputs 24M, test point GPIO0_A6

```
io -4 0x202000fc 0x1f000000 # test clk div set 1
io -4 0x20200144 0x02000000 # enable test clk
io -4 0x202001cc 0x0f000800 # test clk select 24m
io -4 0x20060000 0x30002000 # gpio0_a6 iomux select testclk
```

3.1.6 RV1126

1. Test_clk outputs 24M, test point GPIO1_A4

```
io -4 0xff49022c 0xffff0000 # test clk select 24m and div set 1
io -4 0xff4902dc 0x20000000 # enable test clk
io -4 0xfe010014 0x00070002 # gpio1_a4 iomux select testclk
```

3.1.7 RK3368

1. Test_clk outputs 24M, test point GPIO0_B0

```
io -4 0xff7601bc 0x1f000000 # test clk div set 1
io -4 0xff76021c 0x00020000 # enable test clk
io -4 0xff760380 0x000f0008 # test clk select 24m
io -4 0xff738004 0x00030001 # gpio0_b0 iomux select testclk
```

3.1.8 RK3399

1. Output 24M on Test_clk, test points GPIO2_D1 or GPIO0_B0

```
echo 24000000 > /sys/kernel/debug/clk/clk_testout1_pll_src/clk_rate
echo 24000000 > /sys/kernel/debug/clk/clk_testout1/clk_rate # Set test clk to
24m
echo 1 > /sys/kernel/debug/clk/clk_testout1/clk_enable_count # Enable test clk
io -4 0xff77e004 0x000c0008 # IOMUX select testclk for GPIO2_D1

echo 24000000 > /sys/kernel/debug/clk/clk_testout2_pll_src/clk_rate
echo 24000000 > /sys/kernel/debug/clk/clk_testout2/clk_rate # Set test clk to
24m
echo 1 > /sys/kernel/debug/clk/clk_testout2/clk_enable_count # Enable test clk
io -4 0xff320004 0x00030003 # IOMUX select testclk for GPIO0_B0
```

3.1.9 RK3566/RK3568

Enhanced IO Output CLK Functionality

1. `CLK32K_OUT0` Test Point `gpio0_b0`

```
io -4 0xfdd00100 0x00c00080
io -4 0xfdc20100 0x00010000
io -4 0xfdc20008 0x00030002
```

2. `CLK32K_OUT1` Test Point `gpio2_c6`

```
io -4 0xfdd00100 0x00c00080
io -4 0xfdc20100 0x00010000
io -4 0xfdc20008 0x00030002
io -4 0xfdc60034 0x07000100
```

3. `REF_CLKOUT` Test Point `gpio0_a0`

```
io -4 0xfdc20000 0x00070001
echo 24000000 >/sys/kernel/debug/clk/clk_wifi/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_wifi/clk_enable_count
```

4. `ETH_REFCLK_25M` Test Point `gpio2_c1`

```
io -4 0xfdc60030 0x00700020
echo 25000000 >/sys/kernel/debug/clk/clk_mac0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac0_out/clk_enable_count
```

5. `ETH_REFCLK_25M_M0` Test Point `gpio3_b0`

```
io -4 0xfdc60048 0x00070003
echo 25000000 >/sys/kernel/debug/clk/clk_mac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac1_out/clk_enable_count
```

6. `ETH_REFCLK_25_M1` Test Point `gpio4_b3`

```
io -4 0xfdc60068 0x70003000
echo 25000000 >/sys/kernel/debug/clk/clk_mac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac1_out/clk_enable_count
```

7. `CIF_CLKOUT` Test Point `gpio4_c0`

```
io -4 0xfdc60070 0x00070001
echo 27000000 >/sys/kernel/debug/clk/clk_cif_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cif_out/clk_enable_count
```

8. `CAM_CLKOUT0` Test Point `gpio4_a7`

```
io -4 0xfdc60064 0x70001000
echo 27000000 >/sys/kernel/debug/clk/clk_cam0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cam0_out/clk_enable_count
```

9. `CAM_CLKOUT1` Test Point `gpio4_b0`

```
io -4 0xfdc60068 0x00070001
echo 24000000 >/sys/kernel/debug/clk/clk_cam1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cam1_out/clk_enable_count
```

10. `TESTCLK` Test Point `gpio2_a2`

```
io -4 0xfdc60020 0x07000200
io -4 0xfdd20228 0x1fff0000
io -4 0xfdd20388 0x80000000
```

Note: If you wish to test other frequencies or frequency points, refer to the TRM for detailed instructions by looking up the registers as shown above.

3.1.10 RK3588

Enhanced IO Output CLK Functionality

1. GPIO0_A0 -> refclkout 24M xin_osc0_func

```
io -4 0xfd5f0000 0x000f0001
```

2. GPIO2_C3 -> eth0refclk 50/25M refclko25m_eth0

```
echo 25000000 > /sys/kernel/debug/clk/refclko25m_eth0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_eth0_out/clk_prepare_enable
io -4 0xfd5f8050 0xf0001000
```

3. GPIO4_C3 -> gmac0_clkout 125/50M clk_gmac_50m_cru_i

```
echo 50000000 > /sys/kernel/debug/clk/clk_gmac_50m/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_gmac_50m/clk_prepare_enable
echo 125000000 > /sys/kernel/debug/clk/clk_gmac_125m/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_gmac_125m/clk_prepare_enable
io -4 0xfd5f8090 0xf0001000
```

4. GPIO2_C5 -> 32kout 32K clk_deepslow

```
io -4 0xfd5f8054 0x00f00010
```

5. gpio4_b1 -> clk_mipi_camaraout_m0

```
io -4 0xfd5f8088 0x00f00010
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m0/clk_rate
```

6. gpio1_b6 -> clk_mipi_camaraout_m1

```
io -4 0xfd5f802c 0x0f000200
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m1/clk_rate
```

7. gpio1_b7 -> clk_mipi_camaraout_m2

```
io -4 0xfd5f802c 0xf0002000
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m2/clk_rate
```

8. gpio1_d6 -> clk_mipi_camaraout_m3

```
io -4 0xfd5f803c 0x0f000200
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m3/clk_rate
```

9. gpio1_d7 -> clk_mipi_camaraout_m4

```
io -4 0xfd5f803c 0xf0002000
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m4/clk_rate
```

10. gpio4_b4 -> clk_cifout_out

```
io -4 0xfd5f808c 0x000f0001
echo 24000000 > /sys/kernel/debug/clk/clk_cifout_out/clk_rate
```

11. gpio3_a6 -> eth1refclk 50/25M refclko25m_eth0

```
echo 25000000 > /sys/kernel/debug/clk/refclko25m_eth1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_eth1_out/clk_prepare_enable
io -4 0xfd5f8064 0x0f000100
```

12. gpio4_d5 -> testclkoutm0

litcore 10th frequency division

```
io -4 0xfd818314 0x00ff0009
io -4 0xfd7c0328 0x0e000600
io -4 0xfd818808 0x200000
io -4 0xfd7c0808 0x500000
io -4 0xfd5f809c 0x00f00040
```

3.1.11 RK3528

1. REF_CLK_OUT_M0 Test Point GPIO0_A1

```
io -4 0xFF4B0808 0x00100000
io -4 0xFF540000 0x00f00010
```

2. REF_CLK_OUT_M2 Test Point GPIO3_C3

```
io -4 0xFF4B0808 0x00100000
io -4 0xFF560070 0xf0006000
```

3. CLK_32K_OUT_M0 Test Point GPIO3_C3

```
io -4 0xFF560070 0xf0003000
```

4. CLK_32K_OUT_M1 Test Point GPIO1_C3

```
io -4 0xFF560030 0xf0001000
```

5. TESTCLK Test Point GPIO2_A5

The gpll is divided by 10 and output to testclk.

```
io -4 0xFF570044 0x00f00060
io -4 0xFF4A0394 0x03e00120
io -4 0xFF4A0394 0x3c000000
```

3.1.12 RK3576

1. ref_clk0_out Test Point gpio0a0

```
echo 27000000 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_rate
echo 1 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_prepare_enable
io -4 0x26024000 0x0c000000
io -4 0x26040000 0x000f0001
```

Crystal Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_prepare_enable
io -4 0x26024000 0x0c000400
io -4 0x26040000 0x000f0001
```

Crystal/2 Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_prepare_enable
io -4 0x26024000 0x0c000800
io -4 0x26040000 0x000f0001
```

2. ref_clk1_out Test Point gpio0b4

```
echo 24000000 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_rate
echo 1 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_prepare_enable
io -4 0x26024000 0x30000000
io -4 0x26042000 0x000f0001
```

Crystal Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_prepare_enable
io -4 0x26024000 0x30001000
io -4 0x26042000 0x000f0001
```

Crystal/2 Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_prepare_enable
io -4 0x26024000 0x30002000
io -4 0x26042000 0x000f0001
```

3. ref_clk2_out Test Point gpio0b5

```
echo 50000000 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_rate
echo 1 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_prepare_enable
io -4 0x26024000 0xc0000000
io -4 0x26042000 0x00f00010
```

Crystal Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_prepare_enable
io -4 0x26024000 0xc0004000
io -4 0x26042000 0x00f00010
```

Crystal/2 Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_prepare_enable
io -4 0x26024000 0xc0008000
io -4 0x26042000 0x00f00010
```

4. clk0_32k_out Output Test Point gpio0a2

```
echo 1 > /sys/kernel/debug/clk/xin_osc0_div/clk_prepare_enable
io -4 0x26040000 0x0f000a00
io -4 0x26024000 0x00060000
```

5. ETH_CLK0_25M_OUT_M0 Test Point gpio3a4

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_prepare_enable
io -4 0x26044064 0x000f0003
```

ETH_CLK0_25M_OUT_M1 Test Point gpio2d7

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_prepare_enable
io -4 0x2604405c 0xf0003000
```

6. ETH_CLK1_25M_OUT_M0 Test Point gpio2d6

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_prepare_enable
io -4 0x2604405c 0x0f000200
```

ETH_CLK1_25M_OUT_M1 Test Point gpio1d5

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_prepare_enable
io -4 0x2604403c 0x00f00010
```

7. VI_CIF_CLKOUT Test Point gpio3a2

```
echo 27000000 > /sys/kernel/debug/clk/clk_cifout_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cifout_out/clk_prepare_enable
io -4 0x26044060 0x0f000100
```

8. CAM_CLK0_OUT M0 Test Point gpio3d7

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_prepare_enable
io -4 0x2604407c 0xf0003000
```

CAM_CLK0_OUT M1 Test Point gpio2d2

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_prepare_enable
io -4 0x26044058 0x0f000100
```

9. CAM_CLK1_OUT M0 Test Point gpio4a0

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_prepare_enable
io -4 0x26044080 0x000f0003
```

CAM_CLK1_OUT M1 Test Point gpio2d6

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_prepare_enable
io -4 0x2604405c 0x0f000100
```

10. CAM_CLK2_OUT M0 Test Point gpio4a1

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_prepare_enable
io -4 0x26044080 0x00f00030
```

CAM_CLK2_OUT M1 Test Point gpio2d7

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_prepare_enable
io -4 0x2604405c 0xf0001000
```

11. UFS_REFCLK Test Point gpio4d1

```
echo 26000000 > /sys/kernel/debug/clk/clk_ref_ufs_clkout/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_ref_ufs_clkout/clk_prepare_enable
io -4 0x2722030c 0x00030002
io -4 0x26004b398 0x00f00010
```

12. CLK_TEST_OUT Test Point gpio2a5

gpll 10th Division

```
io -4 0x2720036c 0x00070001
io -4 0x27200368 0xfffc0024
io -4 0x27200808 0x0f000000
io -4 0x26044044 0x00f00050
```

3.1.13 RK2118

1. ref_clk0_out gpio0_b3 1 Division

Crystal bypass

```
io -4 0x50b10308 0x00ff0000
io -4 0x50bc0a00 0x00030002
```

GPLL outputs 40M

```
io -4 0x50b10308 0x00ff0053
io -4 0x50bc0a00 0x00030002
```

V0PLL outputs 49.152M

```
io -4 0x50b10308 0x00ff0097
io -4 0x50bc0a00 0x00030002
```

V1PLL outputs 45.1584M

```
io -4 0x50b10308 0x00ff00d3
io -4 0x50bc0a00 0x00030002
```

2. ref_clk1_out gpio0_a3 1 Division

Crystal bypass

```
io -4 0x50b10308 0xff000000
io -4 0x50bc0000 0xf0001000
```

GPLL outputs 40M

```
io -4 0x50b10308 0xff005300
io -4 0x50bc0000 0xf0001000
```

V0PLL outputs 49.152M

```
io -4 0x50b10308 0xff009700
io -4 0x50bc0000 0xf0001000
```

V1PLL outputs 45.1584M

```
io -4 0x50b10308 0xff00d300
io -4 0x50bc0000 0xf0001000
```

3. ETH_CLK_25M_OUT gpio0_a2

```
io -4 0x50b10304 0xfc007c00
io -4 0x50bc0000 0x0f000100
```

4. OSC_CLK_OUT gpio0_b3 Crystal bypass

```
io -4 0x50bc0a00 0x00030001
```

5. CLK_32K gpio0_a4

```
io -4 0x50B30000 0x00040004
io -4 0x50bc0004 0x000f0001
```

6. CLK_TESTOUT gpio3_c6

xin24 1 Division

```
io -4 0x50b0030c 0x01ff0000
io -4 0x50b00800 0x10000000
io -4 0x501d0074 0x0f000300
```

gpll 10 Division

```
io -4 0x50b0030c 0x01ff0029
io -4 0x50b00800 0x10000000
io -4 0x501d0074 0x0f000300
```

3.1.14 RV1126B/RV1126B-P

1. SAI0_MCLK_M0 gpio7_a2

```
io -4 0x201e00e0 0x0f000100
echo 24576000 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_prepare_enable
```

2. SAI0_MCLK_M1 gpio6_a4

```
io -4 0x201d80c4 0x000f0004
echo 24576000 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_prepare_enable
```

3. SAI1_MCLK_M0 gpio1_b0

```
io -4 0x201b0028 0x000f0002
echo 49152000 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_prepare_enable
```

4. SAI1_MCLK_M1 gpio4_a3

```
io -4 0x201c8080 0xf0002000
echo 49152000 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_prepare_enable
```

5. SAI2_MCLK_M0 gpio3_b6

```
io -4 0x201c006c 0x0f000200
echo 12288000 > /sys/kernel/debug/clk/mclk_sai2_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai2_out2io/clk_prepare_enable
```

6. SAI2_MCLK_M1 gpio5_a3

```
io -4 0x201d00a0 0xf0005000
echo 12288000 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai2_out2io/clk_prepare_enable
```

7. CAM_CLK0_OUT gpio4_b1

```
io -4 0x201c8088 0x00f00030
echo 24000000 > /sys/kernel/debug/clk/clk_mipi0_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi0_out2io/clk_prepare_enable
```

8. CAM_CLK1_OUT gpio4_b0

```
io -4 0x201c8088 0x000f0003
echo 24000000 > /sys/kernel/debug/clk/clk_mipi1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi1_out2io/clk_prepare_enable
```

9. CAM_CLK2_OUT gpio4_a1

```
io -4 0x201c8080 0x00f00030
echo 27000000 > /sys/kernel/debug/clk/clk_mipi2_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi2_out2io/clk_prepare_enable
```

10. CAM_CLK3_OUT gpio4_a0

```
io -4 0x201c8080 0x000f0003
echo 27000000 > /sys/kernel/debug/clk/clk_mipi3_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi3_out2io/clk_prepare_enable
```

11. VI_CIF_CLKOUT_M0 gpio6_c2

```
io -4 0x201d80d0 0x0f000100
echo 27000000 > /sys/kernel/debug/clk/clk_cif_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cif_out2io/clk_prepare_enable
```

12. VI_CIF_CLKOUT_M1 gpio5_c5

```
io -4 0x201d00b4 0x00f00030
echo 27000000 > /sys/kernel/debug/clk/clk_cif_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cif_out2io/clk_prepare_enable
```

13. ETH_CLK_25M_OUT_M0 gpio6_c1

```
io -4 0x201d80d0 0x00f00020
echo 25000000 > /sys/kernel/debug/clk/clk_mac_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac_out2io/clk_prepare_enable
```

14. ETH_CLK_25M_OUT_M1 gpio5_c1

```
io -4 0x201d00b0 0x00f00020
echo 25000000 > /sys/kernel/debug/clk/clk_mac_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac_out2io/clk_prepare_enable
```

15. clk_32k_io 1 Divider Test_clk_out gpio0_a2

```
io -4 0x20040300 0xfff03000
io -4 0x20040800 0x80000000
io -4 0x20040300 0x000f0000
io -4 0x20040800 0x40000000
io -4 0x201a0000 0x0f000200
io -4 0x20130004 0x01000000
```

16. clk_refout 1 Divider Test_clk_out gpio0_a0

```
io -4 0x20040300 0xfff00800
io -4 0x20040800 0x80000000
io -4 0x20040300 0x000f0000
io -4 0x20040800 0x40000000
io -4 0x201a0000 0x000f0002
echo 1 > /sys/kernel/debug/clk/clk_refout/clk_prepare_enable
```

3.1.15 RK2116

1. ref_clk0_out gpio0_a4 1 Division

Crystal bypass

```
io -4 0x40810004 0x000f0001
```

2. ETH_CLK_25M_OUT gpio0_a2

```
io -4 0x40810000 0x0f000100
```

3. CLK_32K gpio0_a1

```
io -4 0x40810000 0x00f00010
io -4 0x40800000 0x00040004
```

4. CLK_TESTOUT gpio2_a3

xin24 1 Division

```
io -4 0x40870310 0x01ff0000
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

gpll 10 Division

```
io -4 0x40870310 0x01ff0029
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

v0pll 10 Division

```
io -4 0x40870310 0x01ff0049
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

v1pll 10 Division

```
io -4 0x40870310 0x01ff0069
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

3.1.16 RK1820

1. CLK_TESTOUT0 gpio0_a0

gpll 10 Division

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80048
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x00070000
io -4 0x1030400000 0x000f0002
```

mpll 10 Division

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80248
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x00070000
io -4 0x1030400000 0x000f0002
```

CLK_32K 1 Division

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80800
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x00070000
io -4 0x1030400000 0x000f0002
```

2. CLK_TESTOUT1 gpio0_a2

dpll 10 Division

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80448
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x1c000000
io -4 0x1030400000 0x0f000200
```

l0pll 10 Division

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80648
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x1c000000
io -4 0x1030400000 0x0f000200
```

3.1.17 RK3538/9

1. REF_CLK_OUT_M0 gpio0_a1

```
io -4 0xfd1b0000 0x00f00010
echo 24000000 > /sys/kernel/debug/clk/clk_refout/clk_rate
echo 1 >/sys/kernel/debug/clk/clk_refout/clk_prepare_enable
```

2. REF_CLK_OUT_M1 gpio5_c2

```
io -4 0xfd2100b0 0x0f000200
echo 49500000 > /sys/kernel/debug/clk/clk_refout/clk_rate
echo 1 >/sys/kernel/debug/clk/clk_refout/clk_prepare_enable
```

3. ETH0_CLK_25M_OUT_M0 gpio0_d5

```
io -4 0xfd1c001c 0x00f00010
echo 25000000 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_rate
echo 1 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_prepare_enable
```

4. ETH0_CLK_25M_OUT_M1 gpio0_b4

```
io -4 0xfd1b000c 0x000f0001
echo 50000000 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_rate
echo 1 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_prepare_enable
```

5. CLK_32K_OUT_M0 gpio3_c2

```
io -4 0xfd1f0070 0x0f000300
```

6. CLK_32K_OUT_M1 gpio5_c2

```
io -4 0xfd2100b0 0x0f000600
```

7. CLK_32K_OUT_M2 gpio0_d4

```
io -4 0xfd1c001c 0x000f0001
```

8. CLK_TEST_OUT_M0 gpio0_a1

sclk_uart1 1 Division

```
io -4 0xfd070308 0xf000b000
io -4 0xfd070808 0x00400000
io -4 0xfd040300 0x0fff0003
io -4 0xfd04080c 0x00070000
io -4 0xfd1b0000 0x00f00020
```

9. CLK_TEST_OUT_M1 gpio2_a5

mclk_sai2 1 Division

```
io -4 0xfd070308 0xf0006000
io -4 0xfd070808 0x00400000
io -4 0xfd060304 0x03ff0100
io -4 0xfd060808 0x03100000
io -4 0xfd1e0044 0x00f00020
```

Note: If you want to test other frequencies or frequency points, refer to the TRM for detailed instructions.

3.2 Software Example

Ensure whether IO supports CLK output (visible in IOMUX functionality), and confirm the supported frequencies for CLK. If it's a TESTCLK function, consult the register to select different functional outputs. For specific functions, simply set the frequency.

3.2.1 RK3288

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT_SRC>;
assigned-clock-parents = <&xin24m>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-rk3288.c b/drivers/clk/rockchip/clk-
rk3288.c
index 2784a7ed05db..a45afa6897a7 100644
--- a/drivers/clk/rockchip/clk-rk3288.c
+++ b/drivers/clk/rockchip/clk-rk3288.c
@@ -204,6 +204,12 @@ PNAME(mux_hsadcout_p)      = { "hsadc_src", "ext_hsadc" };
 PNAME(mux_edp_24m_p)    = { "ext_edp_24m", "xin24m" };
 PNAME(mux_tspout_p)     = { "cpll", "gpll", "npll", "xin27m" };

+PNAME(mux_testout_src_p) = { "aclk_peri", "armclk", "aclk_vio0", "ddrphy",
+                             "aclk_vcodec", "aclk_gpu", "sclk_rga", "aclk_cpu",
+                             "xin24m", "xin27m", "xin32k", "clk_wifi",
+                             "dclk_vop0", "dclk_vop1", "sclk_isp_jpe",
+                             "sclk_isp" };
+
```

```
 PNAME(mux_usbphy480m_p)                 = { "sclk_otgphy1_480m",
"sclk_otgphy2_480m",
                                    "sclk_otgphy0_480m" };
 PNAME(mux_hsicphy480m_p)       = { "cpll", "gpll", "usbphy480m_src" };
@@ -560,6 +566,12 @@ static struct rockchip_clk_branch rk3288_clk_branches[]
__initdata = {
                        RK3288_CLKSEL_CON(2), 0, 6, DFLAGS,
                        RK3288_CLKGATE_CON(2), 7, GFLAGS),

+       MUX(SCLK_TESTOUT_SRC, "sclk_testout_src", mux_testout_src_p, 0,
+           RK3288_MISC_CON, 8, 4, MFLAGS),
+       COMPOSITE_NOMUX(SCLK_TESTOUT, "sclk_testout", "sclk_testout_src", 0,
+                       RK3288_CLKSEL_CON(2), 8, 5, DFLAGS,
+                       RK3288_CLKGATE_CON(4), 15, GFLAGS),
+
        COMPOSITE_NOMUX(SCLK_SARADC, "sclk_saradc", "xin24m", 0,
                        RK3288_CLKSEL_CON(24), 8, 8, DFLAGS,
                        RK3288_CLKGATE_CON(2), 8, GFLAGS),
diff --git a/include/dt-bindings/clock/rk3288-cru.h b/include/dt-
bindings/clock/rk3288-cru.h
index 1f9c62f07389..61ae793438b4 100644
--- a/include/dt-bindings/clock/rk3288-cru.h
+++ b/include/dt-bindings/clock/rk3288-cru.h
@@ -100,6 +100,8 @@
 #define SCLK_MAC_PLL           150
 #define SCLK_MAC               151
 #define SCLK_MACREF_OUT                152
+#define SCLK_TESTOUT_SRC       153
+#define SCLK_TESTOUT           154

 #define DCLK_VOP0              190
 #define DCLK_VOP1              191
```

3.2.2 RK3328

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-rk3328.c b/drivers/clk/rockchip/clk-
rk3328.c
index 1310cfe7798b..5b4ab3360e56 100644
--- a/drivers/clk/rockchip/clk-rk3328.c
+++ b/drivers/clk/rockchip/clk-rk3328.c
@@ -222,6 +222,14 @@ PNAME(mux_mac2io_ext_p)            = { "clk_mac2io",
                                    "gmac_clkin" };
 PNAME(mux_i2s_plls_p)          = { "cpll", "dummy_gpll" };
```

```
+PNAME(mux_sclk_testout_p) =    { "clk_wifi", "dummy", "armclk", "sclk_ddrc",
+                                 "aclk_rkvdec_pre", "aclk_rkvenc",
"aclk_vpu_pre",
+                                 "aclk_rga_pre", "aclk_vio_pre",
"aclk_vop_pre",
+                                 "aclk_gpu_pre", "aclk_bus_pre",
"aclk_peri_pre",
+                                 "aclk_gmac", "dclk_lcdc", "clk_pdm",
"clk_rga",
+                                 "sclk_vdec_core", "sclk_venc_core", "clk_tsp",
+                                 "dummy", "dummy", "dummy", "xin24m"};
+
 static struct rockchip_pll_clock rk3328_pll_clks[] __initdata = {
        [apll] = PLL(pll_rk3328, PLL_APLL, "apll", mux_pll_p,
                     0, RK3328_PLL_CON(0),
@@ -836,6 +844,10 @@ static struct rockchip_clk_branch rk3328_clk_branches[]
__initdata = {
            RK3328_SDMMC_EXT_CON0, 1),
        MMC(SCLK_SDMMC_EXT_SAMPLE, "sdmmc_ext_sample", "clk_sdmmc_ext",
            RK3328_SDMMC_EXT_CON1, 1),
+
+       COMPOSITE_DIV_OFFSET(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+               RK3328_MISC_CON, 8, 5, MFLAGS, RK3328_CLKSEL_CON(2), 0, 5,
DFLAGS,
+               RK3328_CLKGATE_CON(0), 9, GFLAGS),
 };

 static const char *const rk3328_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rk3328-cru.h b/include/dt-
bindings/clock/rk3328-cru.h
index 62479fddb96b..12f205e31273 100644
--- a/include/dt-bindings/clock/rk3328-cru.h
+++ b/include/dt-bindings/clock/rk3328-cru.h
@@ -98,6 +98,7 @@
 #define SCLK_MAC2IO            100
 #define SCLK_MAC2PHY           101
 #define SCLK_MAC2IO_EXT                102
+#define SCLK_TESTOUT           103

 /* dclk gates */
 #define DCLK_LCDC              120
```

3.2.3 PX30

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TEST_OUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TEST_OUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-px30.c b/drivers/clk/rockchip/clk-px30.c
index 9f7a1f91627a..cc4c2fab875d 100644
--- a/drivers/clk/rockchip/clk-px30.c
+++ b/drivers/clk/rockchip/clk-px30.c
@@ -189,6 +189,13 @@ PNAME(mux_wifi_pmu_p)              = { "xin24m",
"clk_wifi_pmu_src" };
 PNAME(mux_uart0_pmu_p)         = { "clk_uart0_pmu_src", "clk_uart0_np5",
"clk_uart0_frac" };
 PNAME(mux_usbphy_ref_p)                = { "xin24m", "clk_ref24m_pmu" };
 PNAME(mux_mipidsiphy_ref_p)    = { "xin24m", "clk_ref24m_pmu" };
+PNAME(mux_sclk_test_out_p)     = { "armclk", "aclk_gpu", "clk_ddrphy4x",
"clk_i2c0", "aclk_vo_pre",
+                                   "aclk_rga", "dclk_vopb", "dclk_vopl",
"aclk_vpu_pre", "aclk_vi_pre",
+                                   "clk_isp", "clk_rtc32k_frac",
"clk_ddrphy1x", "aclk_peri_pre",
+                                   "dummy", "dummy", "dummy", "dummy",
"clk_pwm0",
+                                   "dummy", "aclk_crypto_pre",
"clk_crypto_apk", "xin24m", "aclk_gmac_pre",
+                                   "clk_gmac", "aclk_bus_pre", "clk_pdm",
"clk_i2s0_tx_out", "clk_tsadc",
+                                   "clk_uart1", "clk_saradc", "clk_otp"};

 static struct rockchip_pll_clock px30_pll_clks[] __initdata = {
        [apll] = PLL(pll_rk3328, PLL_APLL, "apll", mux_pll_p,
@@ -907,6 +914,11 @@ static struct rockchip_clk_branch px30_clk_branches[]
__initdata = {
                        PX30_CLKGATE_CON(8), 1, GFLAGS),
        GATE(PCLK_GMAC, "pclk_gmac", "pclk_gmac_pre", 0,
                        PX30_CLKGATE_CON(8), 3, GFLAGS),
+
+       COMPOSITE(SCLK_TEST_OUT, "sclk_test_out", mux_sclk_test_out_p,
CLK_SET_RATE_NO_REPARENT,
+                       PX30_CLKSEL_CON(57), 0, 5, MFLAGS, 8, 4, DFLAGS,
+                       PX30_CLKGATE_CON(16), 15, GFLAGS),
+
 };

 static struct rockchip_clk_branch px30_gpu_src_clk[] __initdata = {
diff --git a/include/dt-bindings/clock/px30-cru.h b/include/dt-
bindings/clock/px30-cru.h
index 644d1f5d26d0..648d7b5ae3a5 100644
--- a/include/dt-bindings/clock/px30-cru.h
+++ b/include/dt-bindings/clock/px30-cru.h
@@ -102,6 +102,7 @@
 #define SCLK_SDMMC_DIV50       87
 #define SCLK_I2S0_TX_MUX       88
 #define SCLK_I2S0_RX_MUX       89
+#define SCLK_TEST_OUT          90
```

3.2.4 RK3308

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-rk3308.c b/drivers/clk/rockchip/clk-
rk3308.c
index 9d09c516bea5..fba208e1a09f 100644
--- a/drivers/clk/rockchip/clk-rk3308.c
+++ b/drivers/clk/rockchip/clk-rk3308.c
@@ -192,6 +192,14 @@ PNAME(mux_spdif_tx_p)              = { "clk_spdif_tx_src",
"clk_spdif_tx_frac", "mclk_i2s0_2
 PNAME(mux_spdif_rx_src_p)      = { "clk_spdif_rx_div", "clk_spdif_rx_div50" };
 PNAME(mux_spdif_rx_p)          = { "clk_spdif_rx_src", "clk_spdif_rx_frac" };
 PNAME(mux_uart_src_p)          = { "xin24m", "usb480m", "dpll", "vpll0",
"vpll1" };
+PNAME(mux_sclk_testout_p) =    { "xin24m", "armclk", "dummy", "dummy",
+                                 "aclk_peri", "hclk_peri", "clk_nandc",
+                                 "clk_sdmmc", "clk_sdio", "clk_emmc",
+                                 "clk_sfc", "dummy", "aclk_bus",
+                                 "hclk_bus", "clk_crypto", "clk_crypto_apk",
"dclk_vop",
+                                 "clk_uart0", "clk_i2c0", "clk_spi0",
+                                 "clk_tsadc", "clk_saradc", "clk_otp",
"hclk_audio"};
+
 static u32 uart_src_mux_idx[]  = { 4, 3, 0, 1, 2 };

 static struct rockchip_pll_clock rk3308_pll_clks[] __initdata = {
@@ -899,6 +907,10 @@ static struct rockchip_clk_branch rk3308_clk_branches[]
__initdata = {
        GATE(PCLK_PWM2, "pclk_pwm2", "pclk_bus", CLK_IGNORE_UNUSED,
RK3308_CLKGATE_CON(7), 13, GFLAGS),
        GATE(PCLK_CAN, "pclk_can", "pclk_bus", CLK_IGNORE_UNUSED,
RK3308_CLKGATE_CON(7), 14, GFLAGS),
        GATE(PCLK_OWIRE, "pclk_owire", "pclk_bus", CLK_IGNORE_UNUSED,
RK3308_CLKGATE_CON(7), 15, GFLAGS),
+
+       COMPOSITE(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+                 RK3308_CLKSEL_CON(73), 8, 5, MFLAGS, 0, 5, DFLAGS,
+                 RK3308_CLKGATE_CON(4), 9, GFLAGS),
 };

 static struct rockchip_clk_branch rk3308_clk_ddrphy[] __initdata = {
diff --git a/include/dt-bindings/clock/rk3308-cru.h b/include/dt-
bindings/clock/rk3308-cru.h
index 5088a0f6fb02..c4707e8775b6 100644
--- a/include/dt-bindings/clock/rk3308-cru.h
+++ b/include/dt-bindings/clock/rk3308-cru.h
```

```
@@ -133,6 +133,7 @@
 #define SCLK_PWM1               120
 #define SCLK_PWM2               121
 #define SCLK_OWIRE              122
+#define SCLK_TESTOUT   123

 /* dclk */
 #define DCLK_VOP                125
```

3.2.5 RV1108

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
--- a/drivers/clk/rockchip/clk-rv1108.c
+++ b/drivers/clk/rockchip/clk-rv1108.c
@@ -159,6 +159,11 @@ PNAME(mux_dclk_hdmiphy_p)  = { "hdmiphy", "xin24m" };
 PNAME(mux_dclk_vop_p)          = { "dclk_hdmiphy", "dclk_vop_src" };
 PNAME(mux_hdmi_cec_src_p)              = { "dpll", "gpll", "xin24m" };
 PNAME(mux_cvbs_src_p)          = { "apll", "io_cvbs_clkin", "hdmiphy", "gpll"
};
+PNAME(mux_sclk_testout_p) =    { "armclk", "aclk_bus_pre", "aclk_vio0",
"aclk_vio1",
+                                 "aclk_periph", "sclk_dsp", "aclk_rkvdec",
+                                 "aclk_rkvenc", "xin24m", "dummy",
+                                 "dclk_vop", "sclk_wifi", "sclk_rga",
+                                 "sclk_isp", "aclk_vpu", "clk_venc_core"};

 static struct rockchip_pll_clock rv1108_pll_clks[] __initdata = {
        [apll] = PLL(pll_rk3399, PLL_APLL, "apll", mux_pll_p, 0,
RV1108_PLL_CON(0),
@@ -776,6 +781,10 @@ static struct rockchip_clk_branch rv1108_clk_branches[]
__initdata = {

        MMC(SCLK_EMMC_DRV,     "emmc_drv",     "sclk_emmc",  RV1108_EMMC_CON0,
1),
        MMC(SCLK_EMMC_SAMPLE,  "emmc_sample",  "sclk_emmc",  RV1108_EMMC_CON1,
1),
+
+       COMPOSITE_DIV_OFFSET(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+                RV1108_MISC_CON, 8, 4, MFLAGS, RV1108_CLKSEL_CON(39), 8, 5,
DFLAGS,
+                RV1108_CLKGATE_CON(9), 9, GFLAGS),
 };
```

```
 static const char *const rv1108_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rv1108-cru.h b/include/dt-
bindings/clock/rv1108-cru.h
index d8d0e0456dc2..b1cbd24a78eb 100644
--- a/include/dt-bindings/clock/rv1108-cru.h
+++ b/include/dt-bindings/clock/rv1108-cru.h
@@ -86,6 +86,7 @@
 #define SCLK_UART0_SRC                 127
 #define SCLK_UART1_SRC                 128
 #define SCLK_UART2_SRC                 129
+#define SCLK_TESTOUT           130

 #define DCLK_VOP_SRC                   185
 #define DCLK_HDMIPHY                   186
```

3.2.6 RV1126

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-rv1126.c b/drivers/clk/rockchip/clk-
rv1126.c
index 05dea9fbd1f0..39905572ca13 100644
--- a/drivers/clk/rockchip/clk-rv1126.c
+++ b/drivers/clk/rockchip/clk-rv1126.c
@@ -220,6 +220,14 @@ PNAME(mux_gpll_cpll_hpll_p)                = { "gpll",
"dummy_cpll", "dummy_hpll" };
 PNAME(mux_gpll_cpll_apll_hpll_p)       = { "gpll", "dummy_cpll", "dummy_apll",
"dummy_hpll" };
 #endif

+PNAME(mux_sclk_testout_p) =    { "xin24m", "clk_rtc32k", "armclk", "dummy",
+                                 "aclk_pdbus", "hclk_pdbus", "clk_core_npu",
+                                 "aclk_pdnpu", "aclk_pdvdec",
"clk_vdec_hevc_ca",
+                                 "aclk_pdvepu", "clk_venc_core", "aclk_pdispp",
+                                 "aclk_pdjpeg", "aclk_pdvi", "aclk_pdvo",
"dclk_vop",
+                                 "clk_spi1", "dummy", "sclk_uart3",
+                                 "aclk_pdphp", "sclk_sfc", "dummy", "dummy"};
+
 static u32 rgmii_mux_idx[]             = { 2, 3, 0, 1 };

 static struct rockchip_pll_clock rv1126_pmu_pll_clks[] __initdata = {
@@ -1378,6 +1386,10 @@ static struct rockchip_clk_branch rv1126_clk_branches[]
__initdata = {
```

```
        GATE(PCLK_DDRPHY, "pclk_ddrphy", "pclk_pdtop", CLK_IGNORE_UNUSED,
                        RV1126_CLKGATE_CON(23), 0, GFLAGS),
 #endif
+
+       COMPOSITE(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+               RV1126_CLKSEL_CON(75), 8, 5, MFLAGS, 0, 5, DFLAGS,
+               RV1126_CLKGATE_CON(23), 13, GFLAGS),
 };

 static const char *const rv1126_cru_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rv1126-cru.h b/include/dt-
bindings/clock/rv1126-cru.h
index 474bcbc546af..2f18e3127113 100644
--- a/include/dt-bindings/clock/rv1126-cru.h
+++ b/include/dt-bindings/clock/rv1126-cru.h
@@ -213,6 +213,7 @@
 #define CLK_NPUPVTM            146
 #define SCLK_DDRCLK            147
 #define CLK_OTP                        148
+#define SCLK_TESTOUT   149

 /* dclk */
 #define DCLK_DECOM             150
```

3.2.7 RK3368

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-rk3368.c b/drivers/clk/rockchip/clk-
rk3368.c
index 009abde551aa..d5a7f720ceb2 100644
--- a/drivers/clk/rockchip/clk-rk3368.c
+++ b/drivers/clk/rockchip/clk-rk3368.c
@@ -157,6 +157,11 @@ PNAME(mux_uart3_p)         = { "uart3_src", "uart3_frac",
"xin24m" };
 PNAME(mux_uart4_p)             = { "uart4_src", "uart4_frac", "xin24m" };
 PNAME(mux_mac_p)               = { "mac_pll_src", "ext_gmac" };
 PNAME(mux_mmc_src_p)           = { "cpll", "gpll", "usbphy_480m", "xin24m" };
+PNAME(mux_sclk_testout_p) =    { "aclk_peri", "armclkb", "aclk_vio0", "dummy",
+                                 "aclk_video", "sclk_gpu_core", "dummy",
+                                 "dummy", "xin24m", "aclk_cci_pre",
+                                 "xin32k", "dummy", "dclk_vop",
+                                 "armclkl", "aclk_gpu_src", "sclk_isp"};
```

```
 static struct rockchip_pll_clock rk3368_pll_clks[] __initdata = {
        [apllb] = PLL(pll_rk3066, PLL_APLLB, "apllb", mux_pll_p, 0,
RK3368_PLL_CON(0),
@@ -881,6 +886,10 @@ static struct rockchip_clk_branch rk3368_clk_branches[]
__initdata = {
        GATE(SCLK_TIMER02, "sclk_timer02", "xin24m", CLK_IGNORE_UNUSED,
RK3368_CLKGATE_CON(24), 2, GFLAGS),
        GATE(SCLK_TIMER01, "sclk_timer01", "xin24m", CLK_IGNORE_UNUSED,
RK3368_CLKGATE_CON(24), 1, GFLAGS),
        GATE(SCLK_TIMER00, "sclk_timer00", "xin24m", CLK_IGNORE_UNUSED,
RK3368_CLKGATE_CON(24), 0, GFLAGS),
+
+       COMPOSITE_DIV_OFFSET(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+                RK3368_MISC_CON, 0, 4, MFLAGS, RK3368_CLKSEL_CON(47), 8, 5,
DFLAGS,
+                RK3368_CLKGATE_CON(7), 1, GFLAGS),
 };

 static const char *const rk3368_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rk3368-cru.h b/include/dt-
bindings/clock/rk3368-cru.h
index 5d3531686790..c6e36b596d24 100644
--- a/include/dt-bindings/clock/rk3368-cru.h
+++ b/include/dt-bindings/clock/rk3368-cru.h
@@ -94,6 +94,7 @@
 #define SCLK_DDRCLK            139
 #define SCLK_TSP               140
 #define SCLK_HSADC_TSP         141
+#define SCLK_TESTOUT   142

 #define DCLK_VOP               190
 #define MCLK_CRYPTO            191
```

3.2.8 RK3399

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTCLKOUT1>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTCLKOUT1>;
clock-names = "soc_24M";
```

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTCLKOUT2>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTCLKOUT2>;
clock-names = "soc_24M";
```

3.2.9 RK3566/8

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&pmucru CLK_WIFI>;
assigned-clock-rates = <24000000>;
clocks = <&pmucru CLK_WIFI>;
clock-names = "soc_24M";
```

3.2.10 RK3588

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru CLK_CIFOUT_OUT>;
assigned-clock-rates = <24000000>;
clocks = <&cru CLK_CIFOUT_OUT>;
clock-names = "soc_24M";
```

3.2.11 RK3528

- CLK_TESTOUT: Not intended for external use at this time, reserved for internal self-test signals.
- CLK_REFOUT: The clock frequency is fixed at 24M by default, can be enabled or disabled, and does not support frequency modification.
- CLK_DEEPSLOW: The clock frequency is fixed at 32K by default, and does not support enabling/disabling or frequency modification.

Thus, only Pinctrl configuration is required.

3.2.12 RK3576

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru CLK_MIPI_CAMERAOUT_M0>;
assigned-clock-rates = <24000000>;
clocks = <&cru CLK_MIPI_CAMERAOUT_M0>;
clock-names = "soc_24M";
```

3.2.13 RK3538/9

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&ref_clkm0_pins>;
assigned-clocks = <&cru CLK_REFOUT>;
assigned-clock-rates = <24000000>;
clocks = <&cru CLK_REFOUT>;
clock-names = "refout";
```

Note:

Control of Pin-ctrll in dts, if in doubt, refer to the Pin-ctrl documentation.
RK3568\RK3588 IO is relatively rich, if you want other IO output CLK, please first check the correspondence between IO and CLK, and then operate according to the example.